Example 11.17

Integrate the following with respect to x:

(i) $\dfrac{e^{2x}-1}{e^{x}}$ (ii) $e^{3x}\left(e^{2x}-1\right)$.

Solution

(i) $$\int \frac{e^{2x}-1}{e^{x}}dx = \int\left(\frac{e^{2x}}{e^{x}}-\frac{1}{e^{x}}\right)dx$$

$$= \int\left(e^{x}-e^{-x}\right)dx = e^{x}+e^{-x}+c.$$

(ii) $$\int e^{3x}\left(e^{2x}-1\right)dx = \int\left(e^{5x}-e^{3x}\right)dx = \frac{e^{5x}}{5}-\frac{e^{3x}}{3}+c.$$

Example 11.18

Evaluate : $\displaystyle\int \frac{1}{\sin^{2}x\cos^{2}x}dx.$

Solution

$$\int \frac{1}{\sin^{2}x\cos^{2}x}dx = \int \frac{\sin^{2}x+\cos^{2}x}{\sin^{2}x\cos^{2}x}dx$$

$$= \int \frac{1}{\cos^{2}x}dx+\int \frac{1}{\sin^{2}x}dx$$

$$= \int \sec^{2}x\,dx+\int \operatorname{cosec}^{2}x\,dx$$

$$= \tan x-\cot x+c.$$

Example 11.19

Evaluate : $\displaystyle\int \frac{\sin x}{1+\sin x}dx.$

Solution

$$\int \frac{\sin x}{1+\sin x}dx = \int\left(\frac{\sin x}{1+\sin x}\right)\left(\frac{1-\sin x}{1-\sin x}\right)dx$$

$$= \int \frac{\sin x-\sin^{2}x}{1-\sin^{2}x}dx = \int \frac{\sin x-\sin^{2}x}{\cos^{2}x}dx = \int \frac{\sin x}{\cos^{2}x}dx-\int \frac{\sin^{2}x}{\cos^{2}x}dx$$

$$= \int \tan x\sec x\,dx-\int \tan^{2}x\,dx$$

$$= \int \tan x\sec x\,dx-\int(\sec^{2}x-1)dx$$

$$= \sec x-\tan x+x+c.$$

Example 11.20

Evaluate : $\displaystyle\int \sqrt{1+\cos 2x}\,dx.$

Solution

$$\int \sqrt{1+\cos 2x}\,dx = \int \sqrt{2\cos^{2}x}\,dx = \sqrt{2}\int \cos x\,dx = \sqrt{2}\sin x+c$$

11_11th-Maths-Vol2-Unit11.indd 198 16-12-2022 11:07:02